

Postfix Expression Puzzle

A web app that shows how a **stack** evaluates postfix (reverse Polish) expressions.

Live demo: *paste your GitHub Pages link here*

DSA concept: Stack (LIFO)

In postfix notation the operator comes after its operands, e.g. `3 4 + 2 *` means `(3 + 4) * 2`. A stack evaluates it with one left-to-right pass:

1. Read a token.
2. If it is a number, **push** it.
3. If it is an operator, **pop** two numbers, apply the operator, and **push** the result.
4. At the end, the single item left is the answer.

Operation	Use in the app	Time
<code>push</code>	Store a number or a result	$O(1)$
<code>pop</code>	Take operands for an operator	$O(1)$
<code>peek</code>	Read the final answer	$O(1)$

The whole expression is evaluated in $O(n)$ time.

Features

- Random puzzles with a check-your-answer box
- Step-by-step stack view with a plain-language explanation of each move
- Run all, restart, and new puzzle buttons
- Load your own expression (numbers with `+ - * )`)

Run it

Open `index.html` in a browser. No install or build step.

Tech

HTML, CSS, and vanilla JavaScript in a single file.